

LIMITS TO STARCH DIGESTION IN THE RUMINANT SMALL INTESTINE^{1,2}

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ABSTRACT

Site and extent of starch digestion by ruminant animals varies with species, grain type and processing method. Based on a review of 40 different experiments with cattle, between 18 and 42% of the dietary starch from corn and sorghum grains fed to cattle reaches the small intestine for digestion. With more extensive grain processing, a smaller quantity of starch reaches the small intestine. In the small intestine, from 47 to 88% of the presented starch is digested, while in the large intestine, 33 to 62% of the presented starch is digested. Though limits to digestion in and absorption from the small intestine can be demonstrated by infusing starch and glucose into the duodenum, enzymatic capacity does not appear to limit intestinal starch digestion since no plateau in the amount of starch disappearing from the small intestine is detected with typical diets. Yet, extent of digestion is incomplete. Other factors, such as time and surface exposure may limit small intestinal digestion of starch. Processing methods to reduce particle size or alter the protein matrix, which cements starch granules together, will increase the extent of digestion both in the rumen and in the small intestine. Performance data from growing cattle fed processed corn and sorghum grains indicate that starch was used 42% more efficiently if it was digested in the small intestine rather than in the rumen. Though total tract starch digestibility is of primary concern, results support the concept that energetic efficiency of growing ruminants is greater if starch is digested in the small intestine rather than in the rumen.

(Key Words: Starch Digestion, Ruminants, Small Intestine, Amylases, Maize, Sorghum.)

Introduction

Starch digestion by ruminant animals has been reviewed and discussed previously (Armstrong and Beever, 1969; Armstrong, 1972; Hale, 1973; Waldo, 1973; Ørskov, 1976). This review is a compilation of more recent research. As companion papers discuss chemical properties of starch, starch digestion in the rumen and grain processing, this paper will examine the factors proposed to limit starch digestion in the small intestine of ruminants. Emphasis will be

placed on high concentrate diets composed of corn and sorghum grains for cattle because starch digestion of other grains and by other species is usually over 99%.

Starch Sources and Site of Digestion

Ruminal and small intestinal digestion (normally measured as disappearance) of starch (α -glucoside) are not fully separable because pre-digestion in the rumen influences both the quantity and composition of starch reaching the small intestine. Starch presented to the small intestine also includes microbial polysaccharide. Bacteria and protozoa contain up to 26 and 38% of their dry matter as polysaccharide (Jouany and Thivend, 1972; Hespell and Bryant, 1979), though values in the 6 to 10% range appear more typical. Microbial polysaccharides can account for all the starch flow to the small intestine with forage diets, but if starch content of duodenal digesta exceeds 10%, as it does with high concentrate diets, some dietary starch must be escaping ruminal fermentation. To determine the site and extent of starch digestion by cattle, results of research trials from 1975 to 1985 from reviewed publications, as well as experiment station reports,

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were compiled. Means for site and extent of starch digestion for cattle fed corn (*Zea mays*) and sorghum (*Sorghum bicolor*) grain diets as classified by grain processing method are summarized in table 1. Total tract digestibility does not necessarily equal the sum of digestion at the individual sites because site of digestion was not measured in all trials. Values for site and extent of digestion discussed below are based on values reported from these experiments. Typically, each experiment had three to five treatments and treatment means were based on three to six animal observations. Values were not weighted by the number of observations within a treatment mean.

Averaged across these 40 cattle trials and 183 treatment means, total tract digestibility of starch averaged only $92.1 \pm 5.5\%$. This is considerably lower than the mean of $99\% \pm 1.2\%$ determined by Waldo (1973). Lower means would be expected from our data base because cattle have consistently lower starch digestion than sheep (Karr et al., 1966), and starch digestion from corn and sorghum is considerably lower than for other grains (Ørskov et al., 1969), probably due to structural differences in the starch as it is stored in the grain.

Ruminal digestion of consumed starch averaged across the 28 trials and 116 treatment means was $72.3 \pm 12.8\%$. Expressed as a proportion of total tract starch digestion, this equals 78.5%, which is identical to the figure calculated by Waldo (1973). This value varies greatly, often being below 65% with limited processing but exceeding 85% for well processed grains. Increased processing generally increased ruminal digestion of starch, which reduced starch flow to the intestine and thereby reduced the percentage of the ingested starch that was available to digest in the small intestine. This is reflected by the simple correlation across processing methods between starch digestion in the rumen, with starch digestion in the small intestine ($r = -.75$; $N = 44$; $P < .01$) and large intestine ($r = -.28$; $N = 44$; $P < .07$) being negative.

These values do not reflect the total capacity for digestion of starch in the small intestine because much of the dietary starch, especially the more easily accessible fraction, has already been digested in the rumen. Even though it is underestimated due to ruminal pre-fermentation of readily available starch, digestive capacity is logically calculated as disappearance only for the quantity of starch that is presented to

the small intestine. Expressed as proportions of the starch entering the small and the large intestines, digestibilities were related in a positive fashion ($r = .20$; $N = 44$; $P < .19$; $r = .26$; $N = 44$; $P < .09$) to extent of ruminal starch digestion. Despite decreasing supply to the intestines, processing tended to increase digestibility of starch in the intestines. This increase could be ascribed to either 1) presence of some quantitative limit to starch digestion in the small intestine so that smaller amounts are more completely digested in the small intestine or 2) the processing methods that increase starch digestion in the rumen also increase starch digestion in the intestines.

To determine whether there is a quantitative limit to digestion of starch in the intestine, disappearance of starch in the small intestine was plotted against input of starch into the small intestine. These two factors increased together ($r = .77$; $N = 44$; $P < .01$). These data (figure 1) show no indication that an increased supply to the small intestine reduced disappearance of starch from the small intestine. When forced through zero, the slope of the regression line was .55. This is reasonably similar to the mean for small intestinal starch digestibility ($52.9 \pm 18.6\%$). Because absorption continued to increase as starch supply increased, these data do not support the suggestion that extent of digestion of starch in the small intestine has a ceiling above which further starch is not digested. Hence, the concept that starch digestibility in the small intestine reaches some quantitative limit under normal feeding practices seems doubtful. This means that the increase in intestinal starch digestion obtained from processing must be due to some effect of processing that altered availability of the starch reaching the small intestine. Nevertheless, digestibility of starch in the small intestine based on only 44 treatment means from 11 different trials averaged $52.9 \pm 18.6\%$. This value indicates that the usefulness of starch that escapes the rumen is relatively low but quite variable. The value is low partly because the more available starch has been removed by ruminal fermentation. If starch were protected from ruminal fermentation and passed to the small intestine, a higher digestibility would be expected. Nevertheless, reasons for this low digestibility deserve attention.

Total post-ruminal starch digestibility based on 112 treatment means from 28 trials averaged $71.7 \pm 17.3\%$. Of the proportion of starch di-

TABLE 1. RECENT MEASUREMENTS OF STARCH DIGESTIBILITY WITH CORN AND SORGHUM GRAINS PROCESSED BY VARIOUS METHODS^a

Processing method	% of starch in diet			% of starch intake			Gain/ feed ^b
	Rumen	Small intestine	Large intestine	Total tract	Small intestine	Large intestine	Post-ruminal
Corn							
Whole	58.9	17.0	2.8	91.7	66.9	33.3	75.4
Cracked	68.9	12.9	8.2	87.6	46.7	55.5	69.9
Rolled	71.8	16.1	4.9	93.2	53.7	37.2	73.1
Ground	77.7	13.7	4.3	93.5	56.3	37.0	66.8
Ensiled	86.0	5.5	1.0	94.6	76.4	55.0	89.8
Steam flaked	82.8	15.6	1.3	97.8	88.1	61.9	87.9
Sorghum							
Rolled	67.8	13.4	5.9	86.4	37.8	33.3	54.8
Ensiled	86.2	9.5	1.1	93.6	69.2	29.5	78.1

^aData from: Cole et al. (1976a,b); Galyean et al. (1976, 1979, 1981); Thornton et al. (1978a,b, 1979); Rust et al. (1979, 1980); Teeter et al. (1979, 1980, 1981); Munifering et al. (1980, 1981); Veira and Macleod (1980); Zinn and Owens (1980b,c); Joanning et al. (1981); Russell et al. (1981); Teeter and Owens (1981); Ferrell et al. (1982); Lee et al. (1982); Rust and Owens (1982a,b); Zinn et al. (1982); Hibberd et al. (1983); Turgeon et al. (1983); Weakley and Owens (1983); Aguirre et al. (1984); Goetsch et al. (1984a,b,c, 1985a,b); Remillard and Johnson (1984); Doran et al. (1985); Kim and Owens (1985).

^bFrom Hale and Prouty (1980).

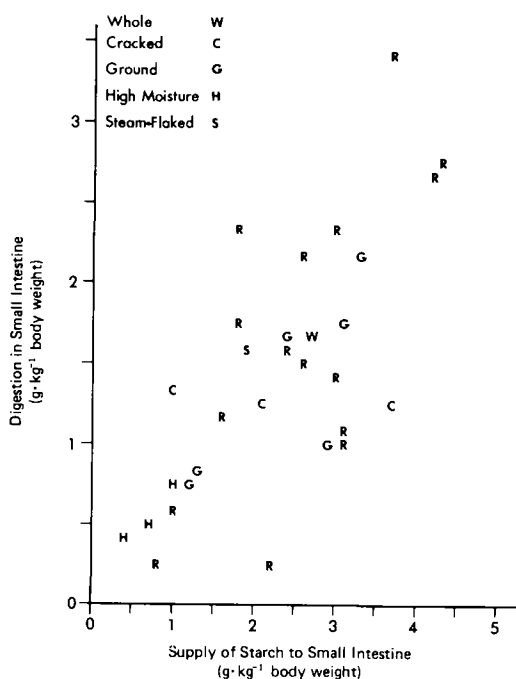


Figure 1. Relationship of starch supply to starch digestion in the small intestine for cattle fed corn grain diets with corn fed in the whole (W), cracked (C), ground (G), high moisture (H) or steam-flaked (S) form. Starch digested = .545 starch supply + .036 ($r^2=.645$; $N=31$; $P<.001$).

gested post-ruminally, about 82% disappeared in the small intestine. Fractional digestion of starch entering the large intestines based on only 44 treatment means from 11 different trials averaged $38.8 \pm 22.1\%$. Fractional digestion in both the small and large intestine increased as total tract digestibility increased ($r=.51$; $P<.01$; $r=.58$; $P<.01$).

Extent of starch digestion both in the rumen and in the total tract were lower for sorghum grain than corn grain processed in a similar manner, as previously reviewed by Waldo (1973). Compared with processing methods that reduce particle size, ensiling generally increased ruminal more than intestinal digestion of starch. To test the impact of animal age (weight), grain level in the diet and level of feed intake on site and extent of digestion, simple correlations of these factors were calculated and are presented in table 2.

Most relationships are not high, do not represent all dependent variables across all independent factors, and only check for linear relationships. Yet these values could be inter-

TABLE 2. SIMPLE CORRELATIONS BETWEEN DIETARY FACTORS AND SITE OF STARCH DIGESTION

Independent factor	Total tract ^a	Rumen ^a	Small intestine ^a	Large intestine ^a	Small intestine ^b	Large intestine ^b	Post-ruminal ^b
Animal wt	.22	.12	-.43	-.07	.31	.47	.30
Level of grain in the diet	-.03	-.10	-.10	-.12	-.15	-.27	-.08
Intake, % of body wt	-.28	-.16	-.03	.10	-.26	-.05	-.05
Starch intake, % of wt	-.23	-.16	-.13	.01	-.30	-.19	-.06

^aStarch digestibility measured as % of starch in diet.

^bStarch digestibility measured as % of starch intake.

preted to suggest that as animal weight (and presumably age) increased, total tract and fractional starch digestion tended to increase despite a drop in the percentage of dietary starch digested in the intestines. As level of grain in the diet increased, total tract starch digestion decreased slightly, with the greatest depression occurring in the large intestine. As total feed or starch intake increased, total tract starch digestion decreased due to reductions in fractional digestion in both the rumen and small intestine.

Factors Limiting Starch Digestion in the Small Intestine

Incomplete intestinal starch digestion was first detected in cleverly designed studies by Larson et al. (1956) and Huber et al. (1961). In the latter study, cattle from 22 to 600 d of age received abomasal infusions of starch at a daily level of 2.2 g/kg body weight. Infused starch failed to elevate blood glucose concentrations to the same degree as infusions of similar amounts of glucose, lactose and maltose. Heating or autoclaving starch increased the blood glucose response. Limits to intestinal starch digestion also have been illustrated by intestinal infusions of pulse doses of starch with measurement of output from the small intestine (Little et al., 1968; Ørskov et al., 1969). Though intriguing, results of these physiological experiments may not apply to normal starch flow to the small intestine because pulse doses often exceeded normal flow, and physical and chemical composition of infused starch (particle size and solubility) differs from that of starch normally escaping ruminal digestion. Doses were high enough to cause diarrhea with infused mono- and di-saccharides but not with amylose. The fact that abomasal infusions of large amounts of glucose caused diarrhea and large quantities of glucose to appear in feces cannot be construed to suggest that under normal feeding conditions dietary glucose is indigestible in the small and large intestine of ruminal animals. Regardless of this limitation, however, such studies proved that starch digestion in the small intestine of cattle is incomplete.

Based largely on these studies, quantitative rather than proportional limits to intestinal starch digestion or absorption were proposed. Specific factors suggested to be responsible included 1) limited activity of amylase, maltase or isomaltase due to inadequate production, inadequate working conditions or presence of

enzyme inhibitors; 2) limited absorption of released glucose from the small intestine; 3) insufficient time for complete starch hydrolysis and 4) inadequate access of enzymes to starch granules due to insolubility or impenetrability of the starch granule. These factors will be discussed individually, although interactions may exist and limiting factors may differ with various grain sources, processing methods and intake levels of feed and starch.

Enzyme Limitations. If enzyme activity limits starch digestion in the small intestine, extent of digestion should decrease as input of starch increases. Results of two early studies support the suggestion that there is a ceiling to digestion of starch in the small intestine. In the pioneering work of Karr et al. (1966) with steers, extent of digestion of starch from ground corn both in the rumen and in the small intestine decreased as corn replaced alfalfa hay in the diet. However, since the diet was changed, differences in intestinal digestion could be ascribed to factors other than starch entry alone. Ørskov et al. (1969) suggested that intestinal starch digestion was limited based on the observation that in cannulated lambs, ileal starch as a percentage of dry matter was equal to .56 times abomasal starch minus .6. This equation, rather than indicating that a ceiling to starch digestion was present, reflects a rather linear relationship of starch disappearance to starch intake with a 56% digestibility. The pulse-dosing studies with purified starch by several researchers (Little et al., 1968; Ørskov et al., 1970) have demonstrated that enzymatic capacity for starch digestion can be exceeded, but as discussed above, level and type of starch infused differs from that normally escaping from the rumen.

Plotted results from grain processing studies show no ceiling to digestion of starch in the small intestine (figure 1). Values are expressed per unit of body weight basis to remove the impact of weight on the regression. Results from these trials illustrate that as starch input to the small intestine increased, disappearance of starch in the small intestine of steers did not plateau or reach a quantitative limit. The maximum value on the graph is for 1,287 g starch reaching the small intestine daily for test steers weighing over 308 kg, which had a small intestinal starch digestibility of 54.6%. Indeed, small intestinal digestion of starch might increase as supply of starch to the small intestine increases if ruminal escape of more readily

digested starch were increased. Because no ceiling to starch digestion was apparent in these studies, enzyme activity does not appear to limit extent of digestion of dietary starch in the small intestine. This does not mean that increasing enzyme activity would be futile, however, since certain other potential limitations such as a rapid passage rate and a large particle size might be partially alleviated by increased enzyme activity. Hence, enzymes involved in starch digestion deserve discussion.

Chemical linkages in amylose and amylopectin were outlined in previous (French, 1973) and companion papers and will not be detailed here. Complete hydrolysis of starch to glucose requires amylase, maltase to cleave released maltose, and isomaltase to cleave the branch points of amylopectin. Bacteria and fungi produce amyloglucosidase, which hydrolyzes starch to glucose directly. Amyloglucosidase is usually employed to hydrolyze starch to glucose for assay of starch content of feeds and digesta.

Amylase. Amylase in the small intestine is produced both by the pancreas and by the intestinal mucosa. Amylase from the pancreas has been extensively studied but amylase from the intestinal mucosa has received limited attention. Pancreatic amylase enters the small intestine via the pancreatic ducts. One of a series of enzymes proportionally synthesized by the pancreas, amylase is stored in pancreatic cells as inactive zymogen granules. Stimulation results in parallel discharge of the various enzymes contained within the zymogen granule by nonruminant animals (Steer and Glazer, 1976). Secretion of pancreatic amylase varies with animal age and diet composition. Siddons (1968) noted that amylase secretion is very low in newborn ruminants, as is true with other mammals, but secretion increases with age. Starch hydrolysis is most rapid when amylase is associated with the intestinal mucosa. Such an association aids transfer of released maltose to the maltase tightly bound to the brush border, which in turn enhances absorption of released glucose.

Diet and intake also can influence enzyme production and secretion. Clary et al. (1969) reported that amylase activity of pancreatic tissue from steers fed an all-concentrate diet was 140% that of pancreatic tissue from similar steers grazing on pasture, and that amylase activity increased linearly with percentage of starch in the diet. Van Hellen et al. (1978) reported that increasing the dietary concentrate

level from 20 to 80% increased both the total flow of pancreatic secretions and amounts of amylase secreted in steers by 1.5- to 2.5-fold.

In addition to total starch intake, percentage starch in the diet can alter amylase secretion. Russell et al. (1981) reported that pancreatic amylase activity was elevated in pancreatic tissue of steers fed higher levels of corn. Hence, long-term feeding of a high grain diet should increase production of pancreatic amylase in ruminants similar to the effect observed in rats (Johnson et al., 1977). With nonruminants, the signal for increased amylase secretion from the pancreas is either blood glucose or insulin. Because blood concentrations of insulin and glucose are generally lower for ruminant than nonruminant animals, stimulation of increased amylase secretion may be more sluggish in ruminant than nonruminant animals (Van Hellen et al., 1978). In nonruminant animals, increased synthesis of amylase in rats could be achieved only when the diet contained high quality protein (Johnson et al., 1977). Whether this was due to an insufficient supply of amino acids for synthesis of the enzyme, or other faults in the adaptive response, is unclear. In some trials (Rust et al., 1979; Veira and Macleod, 1980), added dietary protein has increased total tract digestion of starch by ruminants.

Stability of amylase from the salivary glands of nonruminant animals is dependent upon the presence of Ca, but effects of Ca on stability of pancreatic amylase have not been determined. As calcium chloride and carbonate both increase starch disappearance in the small intestine with low Ca diets (Goetsch et al., 1985a), concentrations of Ca of intestinal digesta could be sub-optimal for amylase stabilization with low Ca diets. Other studies have demonstrated greater response to limestone (CaCO_3) than to an equal amount of Ca from dicalcium phosphate (Turgeon et al., 1983).

The optimal pH for bovine pancreatic amylase activity has been of interest for many years (Armstrong and Beever, 1969; Ørskov et al., 1970). Based on elegant studies by Siddons (1968) and Russell et al. (1981), the optimal pH for pancreatic amylase is near 6.8. A pH decrease or increase by .5 pH units from 6.8 reduced amylase activity by about 20%. Though one could speculate that starch digestion in the small intestine could increase by 20% if intestinal pH were increased from 6.3 to 6.8, this calculation is based on the assumption that

amylase activity limits starch digestion in the small intestine. In steer trials from Colorado, Remillard and Johnson (1984) infused amylase jejunally. Added amylase either with or without added buffer failed to increase starch digestion by steers. To date, no evidence exists that amylase activity limits starch digestion in the small intestine of cattle fed typical diets, despite valiant efforts by many workers to prove this point. In experiments with growing lambs, the capacity for post-ruminal digestion of fed starch is high (Tucker et al., 1968) and amylase activity does not appear to limit starch digestion (Mayes and Ørskov, 1974).

Intestinal pH differs with the method of sampling. Digesta can be obtained repeatedly from cannulated animals or single samples can be obtained from slaughtered animals. Usually the pH of intestinal digesta obtained at slaughter is lower than the pH of digesta from cannulated animals. At slaughter, a peristaltic rush removes much of the digesta from the small and large intestines and whether the residual digesta represents typical chyme is questionable. Intestinal pH also depends on location. In the proximal duodenum, before the point of entry of the pancreatic ducts, pH is usually below 2.5. At this pH, if amylase were present, its activity would be depressed. Addition of bile to chyme will increase its pH. With Holstein steers fed high concentrate diets, pH of the duodenum past the point of entry of the pancreatic ducts was $6.92 \pm .33$ (F. Brzoska, personal communication), quite close to that optimal for amylase action. Near the end of the small intestine, pH is usually neutral or basic. In the terminal ileum, digesta has a strong buffering capacity (Zinn and Owens, 1980a) due to its saturation with bicarbonate.

Intestinal pH also depends on diet composition. The pH of digesta throughout the small intestine is usually lower with concentrate than roughage diets (Russell et al., 1981). This has been attributed to production of acids by microbes in the small intestine (Mayes and Ørskov, 1974; Russell et al., 1981), although concentrations of volatile fatty acids (VFA) and lactate in ileal digesta are typically quite low, and buffering capacity is absent near the pK for VFA. Digesta pH also could be altered by the ion exchange (Na^+ , bicarbonate) which occurs during absorption of glucose (Kimmich, 1981).

The pH of digesta at some locations of the small intestine can fall below the optimum pH

for amylase activity. Dietary buffers have been promoted to increase pH of the digesta, which thereby should increase amylase activity. If amylase activity limits starch digestion, then extent of starch digestion in the small intestine could be increased by addition of buffers. This relationship has been indirectly supported by the results of a number of trials in which addition of limestone to the diet increased both fecal pH and total tract digestion of starch (Russell et al., 1981). Closer examination of this relationship questions whether this relationship is causal or casual.

The theory that added dietary buffers will increase starch digestion is based on two assumptions—that amylase activity limits starch digestion, as discussed earlier, and that dietary buffers will increase pH of the small intestine. Measured directly, starch digestion in the small intestine was not altered by addition of limestone to the diet in one study (Zinn and Owens, 1980c) although in two other studies (Laudert and Matsushima, 1982; Goetsch et al., 1985a) added limestone increased starch digestion in the small intestine but did not increase ileal pH. Although duodenal pH has been elevated slightly by dietary buffers, strong bicarbonate buffering of the jejunal and ileal contents has prevented alteration of pH in these regions of the small intestine (Zinn and Owens, 1980c; Laudert and Matsushima, 1982). Added limestone has never been reported to increase pH obtained from cannulas in the ileum. In one study (Zinn and Owens, 1980a), duodenal digesta was removed and the pH was increased by adding NaOH. Replaced in the small intestine and recovered from the terminal ileum, this chyme at the ileum had a subnormal pH. This indicates that the small intestine internally compensates when duodenal pH is low. Hence, altering pH of the small intestine by adding buffers to the diet, though conceptually valid, appears physiologically infeasible.

Amylase inhibitors also are of concern. Commercially termed “starch blockers,” certain amylase inhibitors have been detected in many legume plants and seeds (Jaffe, 1973) as well as fiber sources (Dunaif and Schneeman, 1981). These compounds and others including tannins and lignin derivatives (Mahmoudzadeh et al., 1985) reduce amylase activity and decrease digestion of starch *in vitro*. Although amylase inhibitors are found in some common feed-stuffs, their ruminal survival, presence and activity in the intestines and potential adapta-

tion by the gut remain to be determined. Although *in vitro* activity is easily measured, *in vivo* action in nonruminants appears limited except at quite high concentrations (Layer et al., 1985). Presence of tannin derivatives adjacent to starch in sorghum grain could presumably deter digestion.

Activity of Other Enzymes. Russell et al. (1981) reported that the optimal pH for pancreatic maltase from steers was 5.8. Increased dietary starch intake did not change maltase activity in any section of the small intestine. Because virtually all of the α -glucoside in the feces of steers was starch and not shorter polymers or maltose, they concluded that intestinal maltase activity does not limit total tract digestion of raw corn starch. Sanford (1982) also suggested that under normal circumstances, the activity of maltase is not rate limiting because monosaccharides, not disaccharides, can be detected in the small intestine. Such accumulation suggests that rate of monosaccharide production exceeded the rate of absorption from, or fermentation within, the small intestine. Because microbes ferment disaccharides such as sucrose and cellobiose within the small intestine (Armstrong and Beever, 1969), microbes also may catabolize maltose and isomaltose though fermentation of monosaccharides would be even more likely. Nevertheless, mere absence of a disaccharide from ileal digesta cannot be considered as proof that appropriate intestinal enzymes are not limiting. Levels of maltase and isomaltase in the ruminant small intestine are usually quite low, although in nonruminants most activity of these enzymes is found along the brush border of the mucosal cells (McMichael and Dahlquist, 1968). Normally, glucose is only released from starch near the gut wall and certain disaccharides may be hydrolyzed inside mucosal cells after absorption. Loss of disaccharidase activity following diarrhea or abrasion of the villi tips has been reported in nonruminants (Plotkin and Isselbacher, 1964). Whether acidic conditions or high fiber levels can cause similar abrasion of the small intestine and loss of disaccharidase activity in ruminants is not known.

Although the majority of starch in common feeds is amylopectin, which contains α -1,6 branch points, the ratio of amylose to amylopectin varies. Varieties of corn are produced whose starch is primarily amylose (high amylose) or primarily amylopectin (waxy). Isomaltase is required to cleave the α -1,6 branch

points for complete hydrolysis of amylopectin. Coombe and Siddons (1973) reported that the pH optimum for bovine isomaltase is 6.0 to 6.2. Isomaltase activity was lowest in the duodenum and higher in the jejunum than in the ileum, which follows the same pattern as maltase. Isomaltase and maltase activities in the small intestine do not increase with age.

Glucose Absorption. Larson et al. (1956) measured blood glucose concentrations to monitor absorption of glucose derived from maltose in the small intestine. Maltose was hydrolyzed and glucose was readily absorbed by cattle 8 to 12 mo of age. Using isolated intestinal sections, they showed that free glucose was readily absorbed from the jejunum and ileum, and also from the cecum, although no free glucose would be expected to be available for absorption from the cecum. Ridges and Singleton (1962) found that the capacity to absorb glucose was considerably lower in adult ruminant than in adult nonruminant animals.

Although glucose absorption is active and extensive by nonruminant animals, feeding pre-ruminant lambs glucose at levels above 10 g/kg body weight, or abomasal infusion of growing lambs more than 3 g/kg body weight daily, causes diarrhea (Walker and Faichney, 1964; Ørskov et al., 1971; Rust, 1983). This observation cannot be extrapolated to suggest that ability to absorb glucose limits starch digestion, however, because digesta normally reaching the small intestine is quite low in soluble carbohydrates due to extensive fermentation in the rumen. Further, infused glucose is in the lumen of the gut, while glucose liberated from polymers is adjacent to the brush border for absorption. Compared with polymerized glucose, free glucose exerts much more osmotic pressure, which will pull water into the lumen and can cause diarrhea.

Huntington et al. (1981) and Huntington (1984) measured glucose uptake from the intestines by determining blood concentrations in peripheral and portal veins. Net transfer of glucose was from the blood to the gut, not vice versa, with diets which consistently have shown disappearance of starch between the duodenum and the ileum. In a more recent study, Huntington and Reynolds (1985) recovered only about 65% of infused glucose and 35% of infused starch in portal blood. The loss could reflect utilization of glucose during absorption by intestinal tissues similar to utilization of amino acids during absorption (Tagari and Bergman,

1978). The observation that glucose appearance from starch was only 54% of that from infused glucose matches well with the suggestion that starch digestion in the small intestine is incomplete, although availability would be expected to be higher for purified starch. An alternative explanation for the observation that disappearance from the gut exceeds portal vein uptake is that microbes ferment soluble sugars in the small intestine before absorption. Presence of microbes in the small intestine is supported by the low but detectable concentrations of cellulase and sucrase (Armstrong and Beever, 1969), and detectable RNA increases (Ben-Ghedalia, 1982) in the small intestine. Bacterial numbers in the small intestine of sheep have been quantitated by Nicoletti et al. (1985). Whether a population of 10^6 organisms per gram of small intestinal tissue is large enough to compete for starch escaping ruminal digestion is uncertain. Incomplete recovery of starch disappearing from the small intestine deserves further study since fermentation in the small intestine would cancel the energetic advantages proposed for starch escaping the rumen (Black, 1971; Ørskov et al., 1971), which will be discussed later.

Time for Digestion and Absorption. Peristaltic contractions push pulse doses of digesta along the small intestine at intervals of 20 to 60 min. Digesta spends less than 3 h in the small intestine of steers (Zinn and Owens, 1980a). Rate of passage of digesta through the small intestine in sheep has been reported to be accelerated by higher levels of feed intake (Gregory et al., 1985). If time for digestion in the small intestine were longer, extent of digestion of starch might be increased.

Consistency of duodenal and ileal digesta varies drastically with diet composition. With alfalfa as a roughage in the diet, ileal digesta is very fluid, possibly due to the high mineral concentrations and osmotic pressures. In contrast, with cottonseed hulls in the diet, ileal digesta is very thick and viscous. High viscosity could limit the exposure of particles or solutes to the intestinal wall, and theoretically could reduce starch digestion in the small intestine. Alternatively, watery digesta dilutes enzymes and reduces their concentration. Digesta flow at the ileum appears to increase with fiber content of digesta (Zinn and Owens, 1980c), so time for intestinal digestion of starch may be altered by fiber source and digestibility.

Accessibility of Starch in Grains. Rate of digestion of starch from grain in the rumen

varies inversely with particle size of the grain (Galvayan et al., 1981). Intact kernels of barley and corn are not attacked in the rumen (Galvayan et al., 1981; Kung et al., 1983). Yet, starch isolated from various grains, regardless of source, is rapidly attacked by ruminal microbes and fermented very rapidly (Hibberd et al., 1983). Hence, non-starch components such as the protein matrix which binds starch granules together, presence of tannin and a thick seed coat, which shelters hard starch from digestion, can limit attack by both microbes and enzymes. Only the larger, more resistant particles survive ruminal attack and pass to the small intestine for digestion. Certain forms of starch such as β -amylose also may resist microbial attack in the rumen.

Particle size also can limit postruminal digestion. Teeter et al. (1980) found that 67 to 75% of the total starch in feces of steers fed whole kernel corn was present in whole corn particles. In another recent study, Rust (1983) placed corn kernels into the duodena of steers and of pigs fed ground grain diets; he then sieved the feces. He recovered virtually all of the grain starch in feces, whether the grain was inserted into the duodenum intact, following pepsin digestion, dehulling, ether extraction or even division into two pieces. Hence, starch that reaches the duodenum in large particles is poorly digested in either the small or large intestine. Many of the processing effects and species differences in starch digestibility can be explained by particle size and starch exposure and alterations due to grain processing and mastication. Younger cattle chew their feed more thoroughly than older cattle, which can explain why younger cattle make more efficient gains than older cattle when both are fed whole shelled corn diets. With rapid eating and high feed intakes, chewing during eating and rumination are limited, and starch passes through the digestive tract with little digestion as seen in Holstein steers in feedlots, which occasionally have feed intakes exceeding 4% of body weight, and poor efficiencies when fed corn in the whole shelled form. Corn grain in diets for dairy cows must be well processed for maximum digestion by dairy cows (Moe et al., 1973; Moe and Tyrrell, 1977), presumably because cows will not chew grain sufficiently.

French (1973) reported that amylase acts most readily on molecular dispersions of starch and only sluggishly, if at all, on the native starch granules. Although amylases of the

animal digestive tract (especially pancreatic α -amylase) can digest uncooked starch, action is slower and less complete than with cooked starch (Walker and Hope, 1963; Mayes and Ørskov, 1974). Hence, dispersing the starch into finer particles should increase digestion in the small intestine as it does in the rumen. Most forms of grain processing will increase access to starch granules, either by physically reducing particle size or acting on endosperm protein to solubilize or denature the gelatinous proteins which cement starch granules together.

Grain Processing

Finely ground or steam-flaked grains are more extensively digested in the rumen than rolled or cracked grain. Hence, supply of starch for digestion in the small intestine is typically greater when the dietary corn or sorghum has a larger particle size. This general relationship has led to the erroneous conclusion that digestion of starch in the small intestine is greatest when grains are fed whole. Though supply of starch to the small intestine usually increases when grains are fed whole, particles reaching the small intestine from whole corn usually are large and are not extensively attacked in the small intestine. As digestibility differences may cancel some of the flow difference, the total amount of starch digested in the small intestine may not be changed markedly by processing. Yet, total tract digestion is generally increased by processing and reducing particle size (Adeeb et al., 1971). This benefit is generally greater with sorghum and corn grain than with other grains (McNeill, 1971), suggesting that access of microbes or enzymes to starch or particle retention in the rumen for digestion varies with grain type.

To be digested in the rumen, large grain particles must be chewed during eating or retained for rumination and re-mastication. When grains are fed with high levels of forage or are treated by certain bases, passage rate of grain through the rumen may be accelerated. This can reduce the time for ruminal digestion and reduce the extent of digestion of starch in the total tract of cattle, despite the increase in rate of digestion in the rumen (Kung et al., 1983). Hence, responses to processing or treatment may differ with processing method and type of grain.

In the large intestine, fermentation of starch is renewed. Ørskov et al. (1970) demonstrated that infusion of purified starch into the cecum of sheep increased both cecal VFA concentra-

tions and fecal N output. When infused, purified starch is rapidly fermented in the large intestine; but, dietary starch that survives to this point has already been subjected to fermentation and digestion. Yet, from 30 to 60% of the starch leaving the small intestine is fermented in the large intestine of cattle.

Fermentation in the large intestine compensates for incomplete fermentation in the rumen to scavenge energy which otherwise would be lost. Large intestinal fermentation is most obvious for cell wall constituents, although site and extent of cell wall digestion can be modified by grain processing. With ground, flaked or high moisture grains, especially at higher levels of feed intake, ruminal pH drops, which inhibits cell wall digestion in the rumen. At high levels of grain intake, ruminal disappearance of acid detergent fiber can approach zero (Zinn and Owens, 1983). By the time digesta reaches the large intestine, however, digestion of starch from flaked grain is nearly complete so that fermentation of cellulose can begin anew. With whole grains, in contrast, ruminal pH usually remains sufficiently high so that digestion of cellulose can proceed albeit slowly. But compensatory digestion in the large intestine with rolled or whole grain diets is usually depressed, probably due to renewed starch fermentation in the large intestine which reduces pH and inhibits digestion of cell wall materials (Aguirre et al., 1984). Hence, grain processing can alter the composition of digesta in the intestines.

Effects of Protein and Calcium Levels and Ionophores

Elevated protein levels have often increased total tract starch digestibility (Rust et al., 1979; Veira et al., 1980). Concentrations of Ca up to .70% of the diet often increase starch digestibility, as well. The protein effect is suggested to occur in the rumen, while Ca may alter digestion in either the rumen or the small or large intestine. Because processing of grain will influence site of digestion of both starch and protein, it also can alter urea utilization. In a series of trials (Gill et al., 1979), an interaction between corn moisture level and source of protein was detected. Urea supplementation resulted in faster and more efficient cattle gains than soybean meal supplementation with dry corn diets. With high moisture corn grain, soybean meal supplements produced slightly more rapid and efficient gains than urea supplements. Later studies by Davis (1982)

indicated that with high moisture corn diets, cattle gains and feed efficiencies were equal whether protein was provided by urea or cottonseed meal. With more extensive ruminal fermentation, usefulness of non-protein N should increase. Although it increases protein solubility greatly, high moisture storage and fermentation of corn grain did not reduce ruminal escape of corn protein (Aguirre et al., 1984).

Although added bases can increase rate of starch digestion from whole grains in the rumen, in some trials added bases have decreased total tract digestibility of grain due to faster passage through the rumen or possibly localized adverse effects on ruminal bacteria. Ionophores have usually increased total tract starch digestibility of less well processed grains (Rust et al., 1979; Ferrell et al., 1982) although in one report, monensin decreased ruminal digestion of starch (Muntifering et al., 1981). Growth stimulants or other compounds that increase feed intake would be expected to decrease slightly total tract starch digestibility of less well processed grains, although increased feed intakes of high concentrate diets may not greatly reduce extent of ruminal digestion of starch (Zinn and Owens, 1983). Type and level of roughage in the diet can have opposing effects on starch digestion. If added roughage increases mastication and rumination of the grain, it should increase digestibility in both the rumen and small intestine, but if added roughage elevates passage rate and more large particles are flushed out of the rumen, ruminal and total tract digestibility of starch from less well processed grains will be compromised (Teeter et al., 1981). The interaction between grain processing and roughage level is most apparent with diets of whole shelled corn. With a whole shelled corn diet, levels of roughage above 6% depress starch digestion in the total tract and depress efficiency of feed use. In contrast, with a flaked corn diet, roughage levels as low as 6% result in poor efficiency, possibly due to ruminal acidosis. Further research of these interactions between grain processing and source and level of roughage are needed to optimize the efficiency with which grain can be used by ruminant animals.

Impact of Site of Starch Digestion on Energetic Efficiency

Infusions of glucose into the rumen or the small intestine have demonstrated that effi-

ciency of utilization of energy is greater if carbohydrate (glucose) is absorbed from the small intestine than if it is fermented in the rumen (Armstrong et al., 1960). Yet, efficiency of feed use by cattle generally favors more extensive processing methods, which increase extent of ruminal digestion (Hale, 1973). This dilemma has been of concern to many researchers. Because starch availability differs from glucose availability, one cannot directly extrapolate from glucose to starch to appraise the relative value of digestion at the two locations. Further, estimating energy value of grain processed by various methods is complicated by the fact that processing alters ruminal and total tract digestion as well as the site of digestion.

To address this problem more directly, multiple regression was used to determine the effects of extent of digestion of starch 1) in the rumen and 2) in the small intestine on gain to feed ratio for sorghum and corn grains processed by various methods. Gain-to-feed ratios as presented in table 1 were obtained from an extensive review of grain processing trials by Hale and Prouty (1980), while values for site of digestion were means from table 1.

When forced to a zero-intercept, efficiency of feed use (gain-to-feed ratio) was equal to .159 times ruminal starch digestion (% of fed) plus .227 times small intestinal starch digestion (prediction $r^2 = .60$; $SE_{y-x} = .006$). Although imprecise due to the inaccuracy of input factors, these values indicate that increasing starch digestion in either the rumen or small intestine will increase the gain-to-feed ratio. But the magnitude of the coefficients indicate that starch digested in the rumen is used only 70% as efficiently as starch digested in the small intestine for gain by steers. This value is reasonably similar to values determined by ruminal vs abomasal infusion of glucose (69 to 76%; Armstrong et al., 1960), and calculated (62 to 82% as much net energy for production; Black, 1971) from differences in loss of energy as methane, heat of fermentation and digestion and excreta. Hence, as source of energy, starch digested in the small intestine provides 42% more energy than starch digested in the rumen. Reducing digestion in the rumen, however, reduces the amount of energy available for ruminal microbes and may decrease the supply of microbial protein available for the animal. Of least value is starch digestion in the large intestine, which increases fecal N loss and decreases apparent protein digestibility (Ørskov et al., 1970).

In a trial to test the theory that absorption of glucose from the small intestine is more energetically efficient than digestion to VFA in the rumen, Rust (1983) provided 180 g glucose daily to 21-kg lambs either in the feed or via abomasal infusion for 165 d. Energy retention in the empty body from fed glucose was only 52% of that from abomasally infused glucose. Surprisingly, 87% of the increased energy retained was as lipid, primarily around the intestines and in the omentum. Greater lipid deposition with abomasal than ruminal glucose infusion also was reported by Armstrong et al. (1960). Hence, the potential for hormonal changes and lipogenesis by intestinal tissue, as well as limitations in transport of glucose from the site of absorption, need further research attention.

Since processing increases total tract digestion, processing usually improves feed efficiency. For maximum efficiency, digestion in the small intestine may be preferred. But processing to increase digestibility of starch reaching the small intestine usually increases ruminal digestion, as well. Feeding particles in a coarse form to increase flow of starch to the small intestine appears infeasible, however, due to the sacrifice in total tract digestibility of starch. Other methods to shift site of starch digestion to the small intestine deserve attention. These include modulating the rate of starch digestion in the rumen with antimicrobial or chemical compounds, or altering passage rate from the rumen with agents which change volume or motility of the rumen, salivary input, or size or location of particles in the rumen. Because the smallest starch particles are most rapidly digested in the rumen, and very large particles pass through the intestines unscathed, processing to a particle size larger than 250 μm but smaller than 1,000 μm may be ideal (Kim and Owens, 1985). Such a size should help avoid ruminal acidosis problems and avoid post-ruminal depressions in digestibility. Both the mean and the distribution of particle sizes appear important and could explain how two processing methods which produce similar mean particle sizes—rolling and grinding—result in different ruminal and postruminal starch digestibilities.

In conclusion, starch digestion in the total digestive tract is usually over 95%, although degree of processing alters this figure greatly. Ruminal and total tract digestion is high for most grains and most species except for corn

and sorghum grain for large ruminants, especially cattle. Digestibility coefficients rank processing methods, from least to greatest, as whole, rolled, ground, high moisture and steam flaked grains. Of total tract starch digestion of corn and sorghum by cattle, an average of 80% occurs in the rumen. Of starch reaching the intestines in cattle, only about 65% is digested therein, with an average of 82% of the disappearance occurring in the small intestine. Processing grain to increase total tract digestion usually increases not only the extent of digestion of starch in the rumen but also the digestibility of starch which reaches the small intestine. Though enzyme limitations could potentially limit digestion of starch in the small intestine under certain conditions, with typical diets the physical size of particles containing starch appears more frequently to limit digestion of starch in the intestines. With higher roughage diets, starch digestion tends to be shifted downstream, but higher starch intakes have not consistently reduced digestion in the rumen or in the small intestine. Though total tract digestibility of starch is usually very high, its site of digestion can indirectly alter the site and extent of digestion of cell wall components. These interactions can explain a portion of the negative associative effects between grains and forages. Starch digested in the small intestine has 42% more value than starch digested in the rumen. Hence, if starch that escapes the rumen can be digested to an extent exceeding 70% of its digestibility in the rumen, starch escape from the rumen should improve energetic efficiency of production by ruminant animals.

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